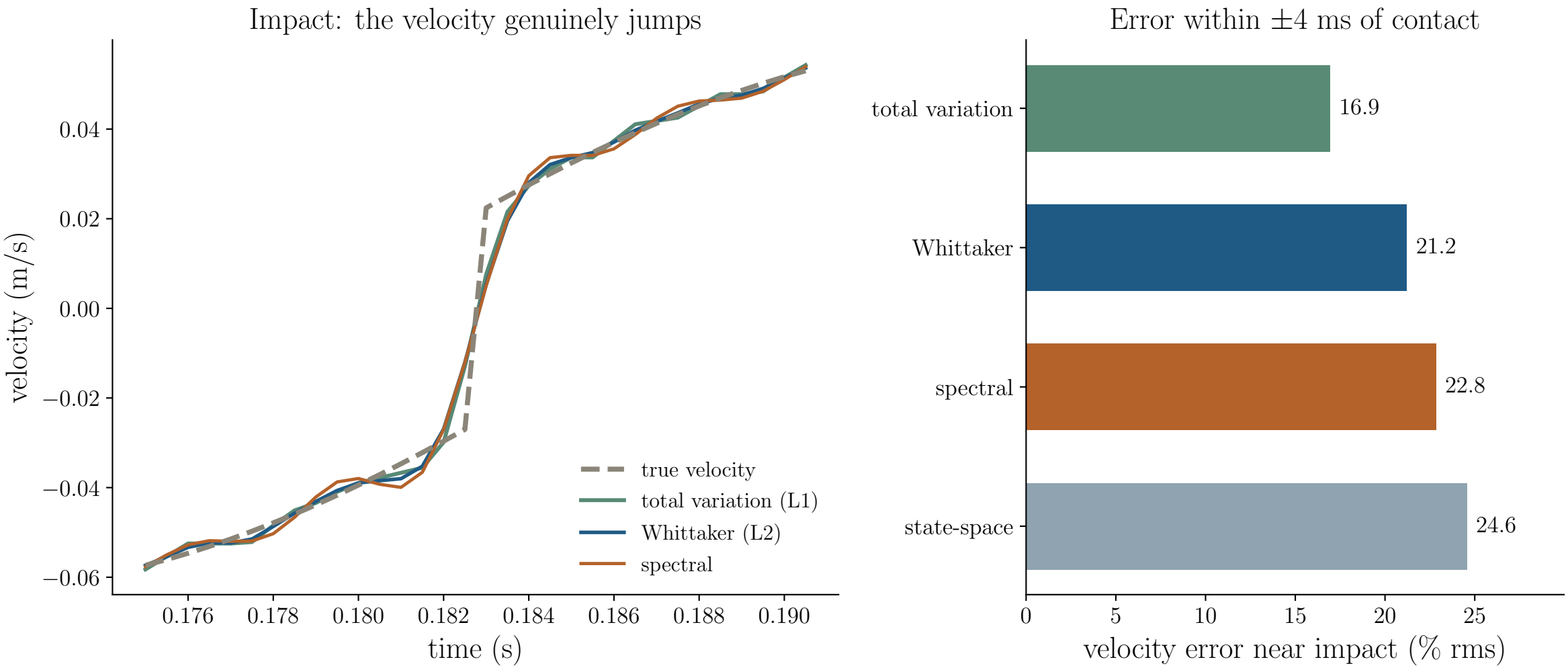
Impact: the velocity genuinely jumps
velocity (m/s)
0.04
0.02
0.00
−0.02
−0.04
−0.06
0.176
0.178
0.180
0.182
0.184
0.186
0.188
0.190
time (s)
true velocity
total variation (L1)
Whittaker (L2)
spectral
Error within ±4 ms of contact
total variation
Whittaker
spectral
state-space
16.9
21.2
22.8
24.6
0
5
10
15
20
25
velocity error near impact (% rms)